

Unlike the Android and Windows Phone emulators, the iOS Simulator does not run in an emulated or virtualised environment. Instead it simulates the software environment of the iPhone. The simulator includes basic iOS apps and functionality but compiled to run on an OSX desktop. When an app is simulated in this environment, it too is compiled to run on the desktop platform as an OSX app using features from the iOS platform. This also means that such apps have access to the full power of the OS, the entire RAM and CPU power of the computer it



Firefox OS Simulator running on Linux.

is running on. The architecture of the hardware that the Simulator runs on is very different compared to the architecture of the actual phone, as the Simulator runs on an x86 Intel chip while the mobile architecture is ARM. The iOS simulator simulates the the software environment in which your application will run, while the hardware is entirely different.

Another mobile OS simulator is the Firefox OS simulator. Firefox OS is essentially a very complex web application that manages other web applications. All parts of the user-visible OS are running on a browser that is running directly on the device. This means you can simply run Firefox OS on a supported browser, and that is just what the Firefox OS Simulator does. It is a addon for Firefox that includes the entire OS code. Like the iOS Simulator this app is running directly on your computer and as such its performance will reflect that of your computer, and not the actual device.